Introduction to Quarto

Write, analyze, and publish in a single document

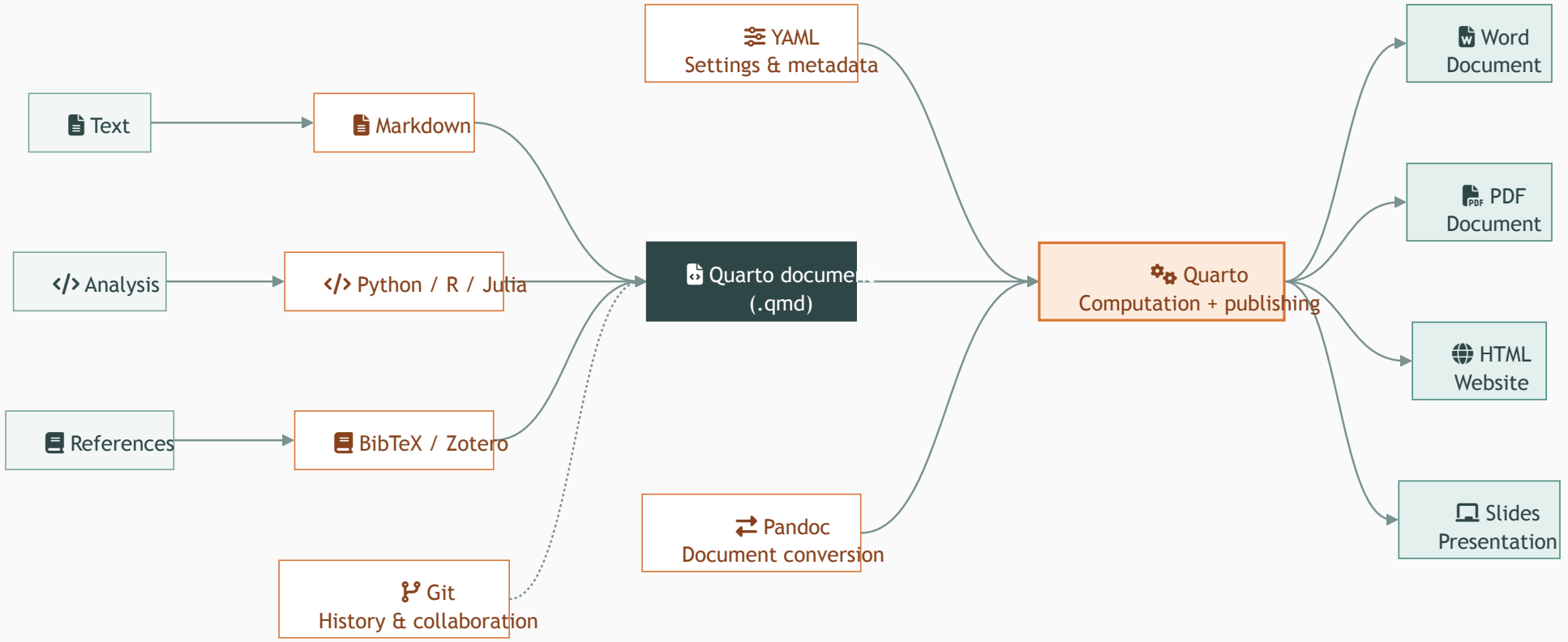
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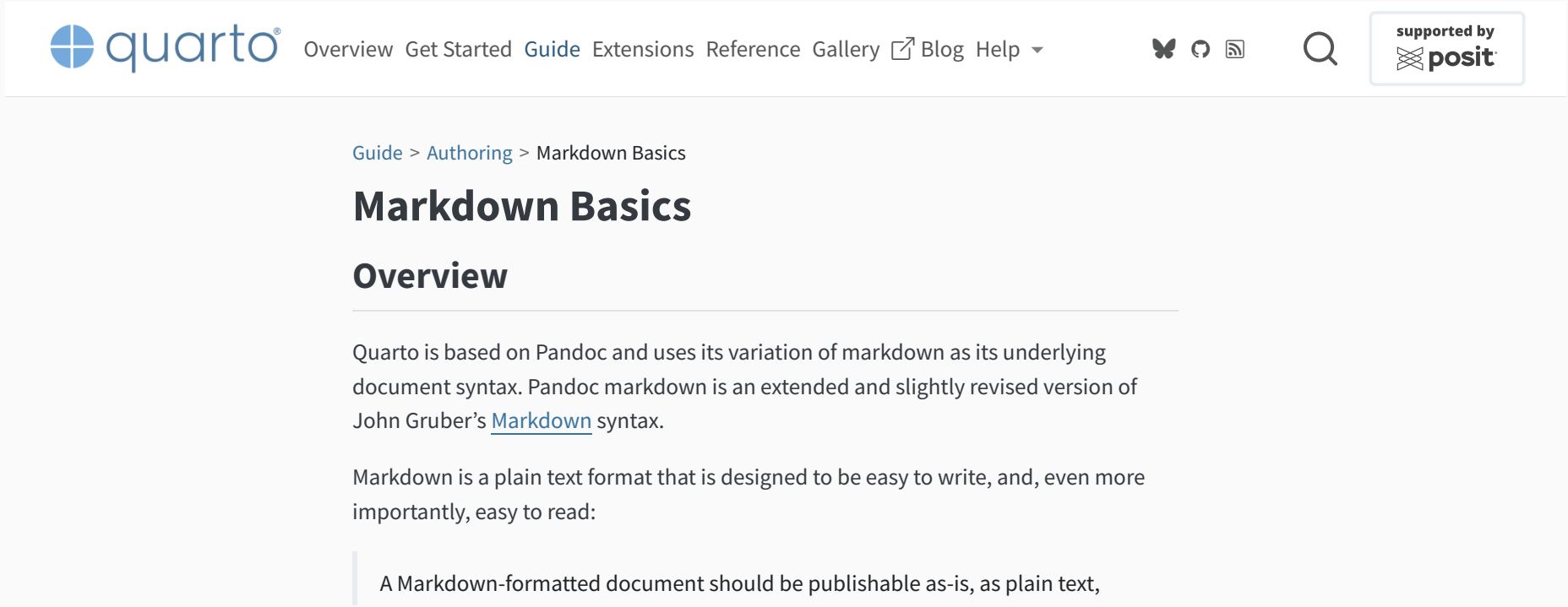
Why Quarto?

Why Quarto


YAML
Settings & metadata
Text
Markdown
Word
Document
PDF
Document
Analysis
Python / R / Julia
Quarto
Computation + publishing
HTML
Website
References
BibTeX / Zotero
Pandoc
Document conversion
Slides
Presentation
Git
History & collaboration

Markdown

- What? Lightweight markup for headings, lists, links and emphasis.
- Why? Keep the source readable and focus on content and structure.



Open Quarto Markdown Basics ↗

Pandoc

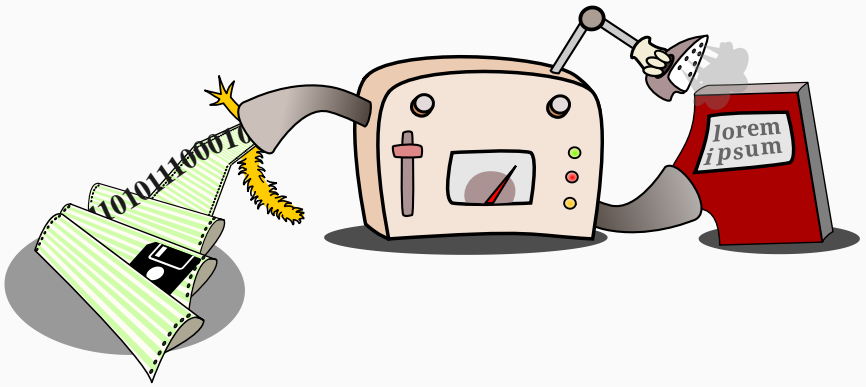
- What? Converts documents between formats such as Markdown, HTML and Word.
- Why? One source, many outputs: the conversion machinery inside Quarto.

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Pandoc a universal document converter

About Installing Demos ▼ Documentation Help Extras Releases

Search



If you need to convert files from one markup format into another, pandoc is your swiss-army knife. Pandoc can convert between the following formats:

(← = conversion from; → = conversion to; ↔ = conversion from and to)

Lightweight markup formats

Open the Pandoc website ↗

YAML

- What? A readable data format built from keys, values and indentation.
- Why? In Quarto: declare the document's metadata and publishing settings.

About YAML

What is YAML?

YAML (YAML Ain't Markup Language) is a human-readable data serialization language designed for data exchange between languages with different data structures.

It's often used for configuration files, data exchange, and representing complex data structures in a readable format.

Key Features

- **Human Readable** - Easy to read and write for both humans and machines
- **Language Independent** - Works across programming languages and platforr
- **Hierarchical** - Supports complex nested data structures
- **Extensible** - Can be extended with custom data types
- **Secure** - Built with security considerations in mind

Open the YAML website ↗

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Git

- What? A distributed version control system that records changes to files.
- Why? Keep a history, compare versions, and collaborate on shared work.

git --print-out

Git is a free and open source distributed version control system designed to handle everything from small to very large projects with speed and efficiency.

Git is lightning fast and has a huge ecosystem of GUIs, hosting services, and command-line tools.



About
Git's performance and ecosystem

Learn
Pro Git book, videos, tutorials, and cheat sheet

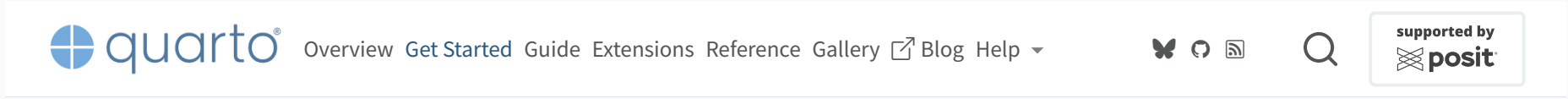
Latest source release
2.56.0
Release Notes (2026-09-28)

Open the Git website ↗

What is Quarto

- **Open-source scientific and technical publishing system**
  → Leverages modern tools for content and structure (Markdown) and configuration (YAML)
- **Favours reproducibility and transparency**
  → Supports executable code within text
- **Supports multiple output formats**
  → Pandoc is the conversion machinery underneath!
- **Successor to R Markdown but language-agnostic**
  → *Python*, *R*, *Julia*, ...
- **Integration in most IDE**
  → *VS Code*, *RStudio*, *Jupyter*...

Getting started



Get Started

Install Quarto, then check out the tutorials to learn the basics.

Step 1

Install Quarto

Download Quarto CLI

1.11.9 (Mac OS) | Date: Oct 30, 2026

Platform	Download	Size	SHA-256
Ubuntu 18+/Debian 10+	quarto-1.11.9-linux-amd64.deb	134.26 MB	`64078d1…`
Linux Arm64	quarto-1.11.9-linux-arm64.deb	134.6 MB	`17aef0c…`
Mac OS	quarto-1.11.9-macos.pkg	226 MB	`35ad09e…`
Windows	quarto-1.11.9-win.msi	134.13 MB	`d0c0ac3…`

Release notes and more downloads…

Objectives

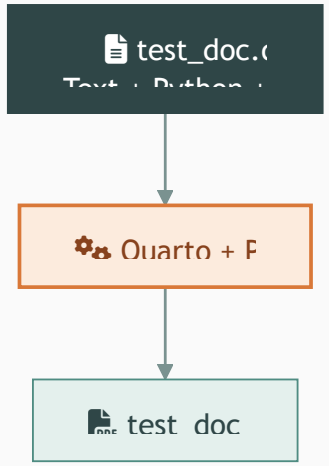
Aim: **Demonstrate the "Write, Analyze, Publish" workflow**

- **Part I**: Quarto for research
  - → Write a document from scratch
  - → Write a document with a more complex workflow
  - → Complex multi-output workflow
- **Part II**: Quarto for teaching
  - → Write a textbook website
  - → Add interactive exercises
  - → Develop interactive slides
  - → Online publication on Github pages

Quarto for research

A first document from scratch

- **One document** containing:
  - → Text (→ *Markdown*)
  - → Code (→ *Python*)
  - → Configuration (→ *YAML*)
- **One output** → HTML
- Good strategy for **basic workflows**

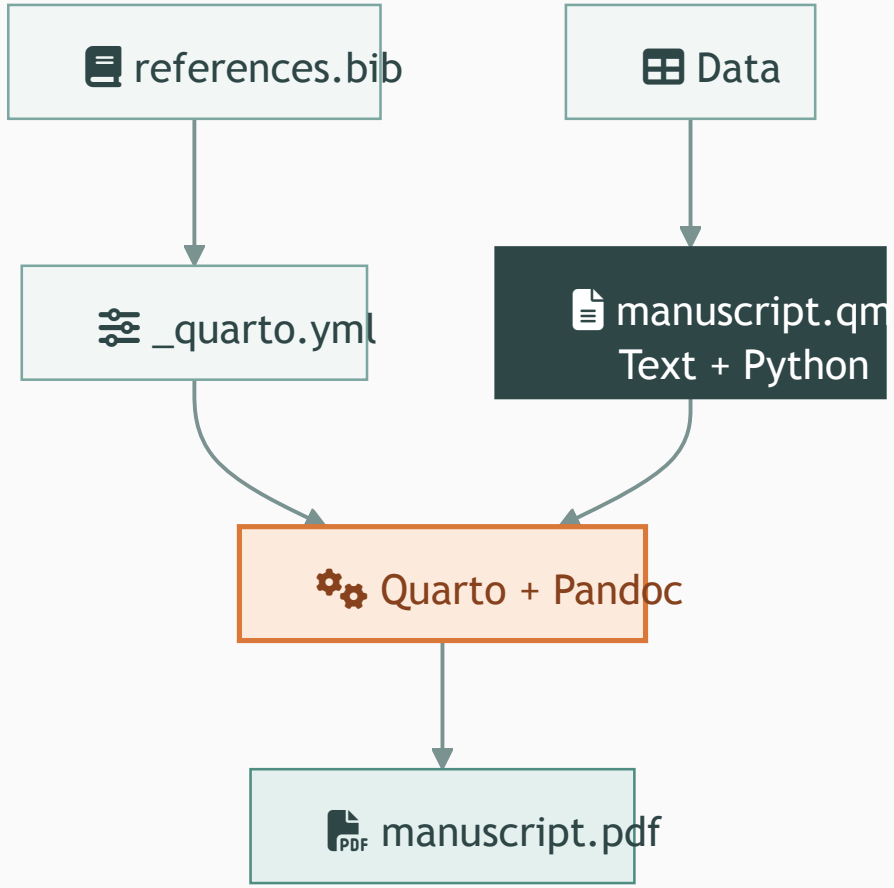



```
quarto preview test_doc.md
quarto render test_doc.md
```

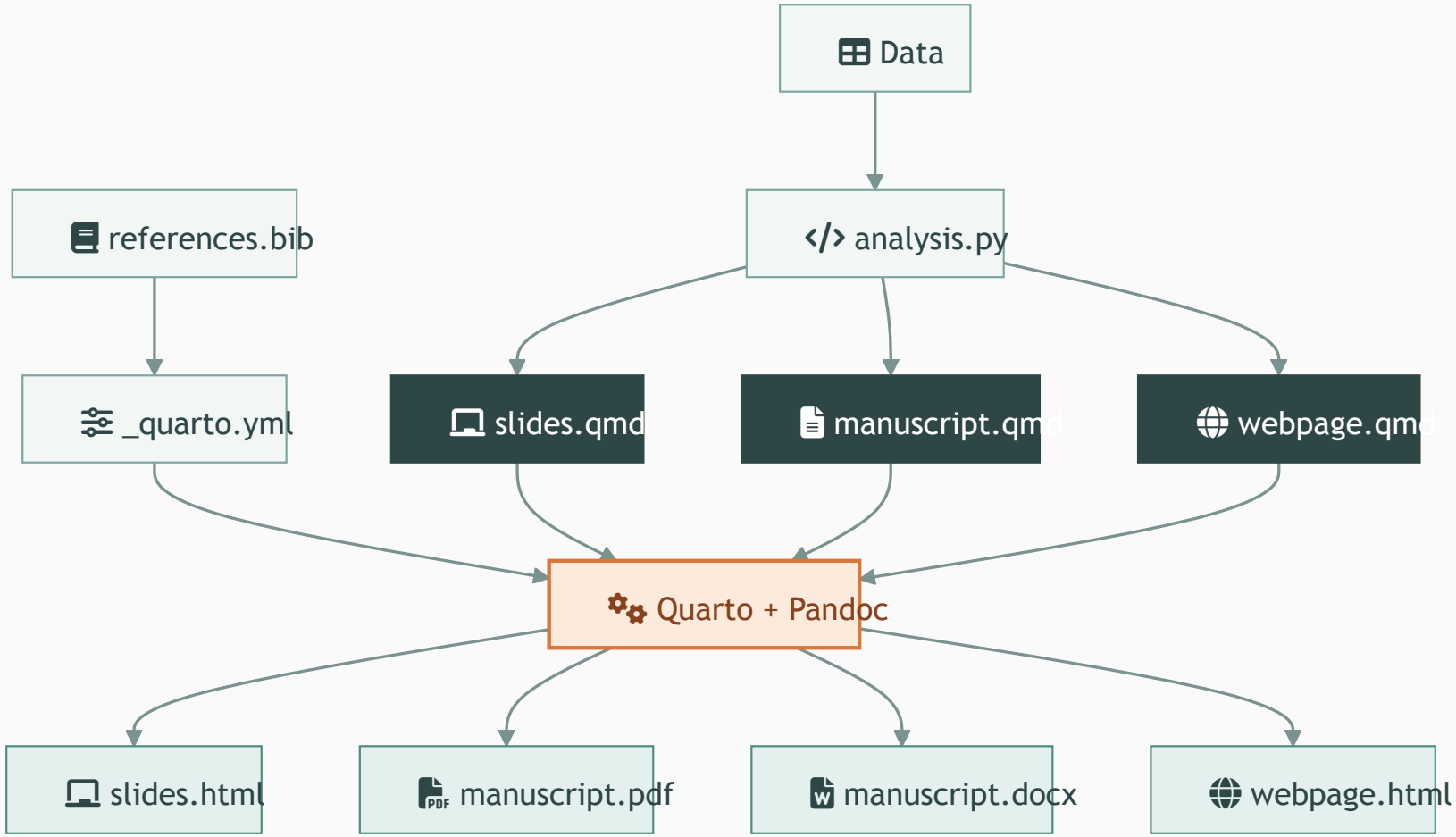
One document, one PDF

- **More complex workflow**
- **Separated**
  - → Configuration
  - → Dataset
  - → Bibliography
- **One output** → PDF

`quarto preview` `quarto render`



One analysis, several documents



Quarto for teaching

Text book

- **Teaching book**
  - → Cross-referencing
  - → Collaborative tools
- **Exercises**
  - → Profiles for *students*, *lecturer* or *corrections*
  - → Interactive coding exercises
- **Slides**
  - → Drawing
  - → Code integration + animation
  - → Live coding
  - → Integration with medias
  - → Other cool tools

Now what?

Conclusion

... easy to get started

... learning curve to get into some more complex workflows

This presentation / examples

My teaching template

Examples

Volcanology

Outils informatiques

Data science